MLP: First layer only

All layers trainable; LR changes at 448; same warm Adam state and all 5,000 training examples.

Unchanged Adam (LR 0.001) — 0.5× intervention — 0.25× intervention

I9: 0.5× LR
First layer 0.0005; later layers 0.001

I10: 0.25× LR
First layer 0.00025; later layers 0.001

(a) Full training loss
(b) $\eta\hat{\kappa}_t$ (symlog)
(c) Raw-gradient cosine
(d) Full $\|g_t\|$ (log)
(e) Later-layer $\|g_t\|$ (log)
(f) First-hidden change cosine
(g) First-hidden RMS (log)
(h) Output-change cosine
(i) Output RMS (log)
(j) Applied learning rate

Completed optimizer updates t

Loss: independent linear axes. Other rows share scales across all pages. No smoothing.
Row (c): solid = full raw gradient; dashed = later-layer gradient. Row (j): solid = first-layer LR; dashed = later-layer LR.
First-hidden and output metrics retain all sample coordinates. Changes are outgoing. $\eta\hat{\kappa}_t$ is a projected-gradient diagnostic under Adam.

MLP: All layers: global LR controls

All layers trainable; LR changes at 448; same warm Adam state and all 5,000 training examples.

Unchanged Adam (LR 0.001) — 0.5× intervention — 0.25× intervention

I11: 0.5× LR
First layer 0.0005; later layers 0.0005

I12: 0.25× LR
First layer 0.00025; later layers 0.00025

(a) Full training loss
(b) $\eta\hat{\kappa}_t$ (symlog)
(c) Raw-gradient cosine
(d) Full $\|g_t\|$ (log)
(e) Later-layer $\|g_t\|$ (log)
(f) First-hidden change cosine
(g) First-hidden RMS (log)
(h) Output-change cosine
(i) Output RMS (log)
(j) Applied learning rate

Completed optimizer updates t

Loss: independent linear axes. Other rows share scales across all pages. No smoothing.
Row (c): solid = full raw gradient; dashed = later-layer gradient. Row (j): solid = first-layer LR; dashed = later-layer LR.
First-hidden and output metrics retain all sample coordinates. Changes are outgoing. $\eta\hat{\kappa}_t$ is a projected-gradient diagnostic under Adam.

MLP: Later layers only: location controls

All layers trainable; LR changes at 448; same warm Adam state and all 5,000 training examples.

Unchanged Adam (LR 0.001) — 0.5× intervention — 0.25× intervention

I13: 0.5× LR
First layer 0.001; later layers 0.0005

I14: 0.25× LR
First layer 0.001; later layers 0.00025

(a) Full training loss
(b) $\eta\hat{\kappa}_t$ (symlog)
(c) Raw-gradient cosine
(d) Full $\|g_t\|$ (log)
(e) Later-layer $\|g_t\|$ (log)
(f) First-hidden change cosine
(g) First-hidden RMS (log)
(h) Output-change cosine
(i) Output RMS (log)
(j) Applied learning rate

Completed optimizer updates t

Loss: independent linear axes. Other rows share scales across all pages. No smoothing.
Row (c): solid = full raw gradient; dashed = later-layer gradient. Row (j): solid = first-layer LR; dashed = later-layer LR.
First-hidden and output metrics retain all sample coordinates. Changes are outgoing. $\eta\hat{\kappa}_t$ is a projected-gradient diagnostic under Adam.